

4th Place Solution

Rank	4
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Topic ID	611893
Version	Original

Source: <https://www.kaggle.com/competitions/rsna-intracranial-aneurysm-detection/discussion/611893>

Retrieved with Kaggle CLI on 2026-07-07.

Sincere Thank You to RSNA and Kaggle for hosting this competition

I tackled the challenge as a speedrun as I had only 14 days left on the competition when I started, which is nowhere near enough to cover everything one can do with this dataset, the dataset is super cool and I have barely scratched the surface. I managed to get in the gold medal line in 8 days.

Finally a GM, yay!

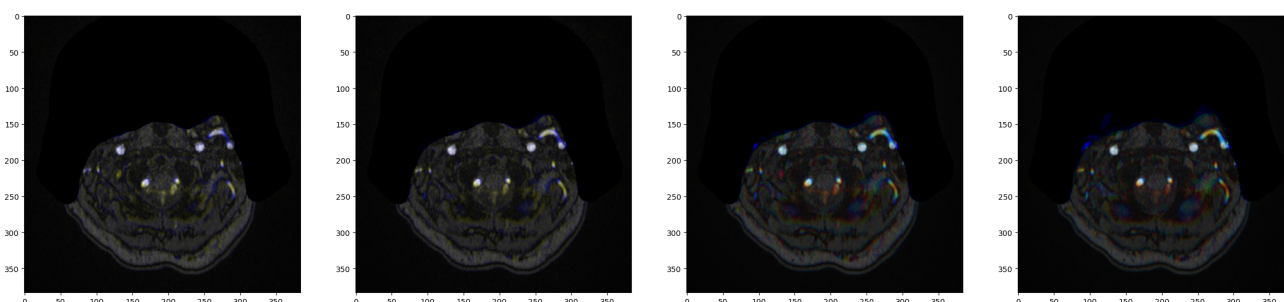
Segmentation Model Failure

I started with training a segmentation model... which was not really happy to learn beyond 0.6 Dice in the short time I tried which was nowhere near enough for my confidence in the approach.

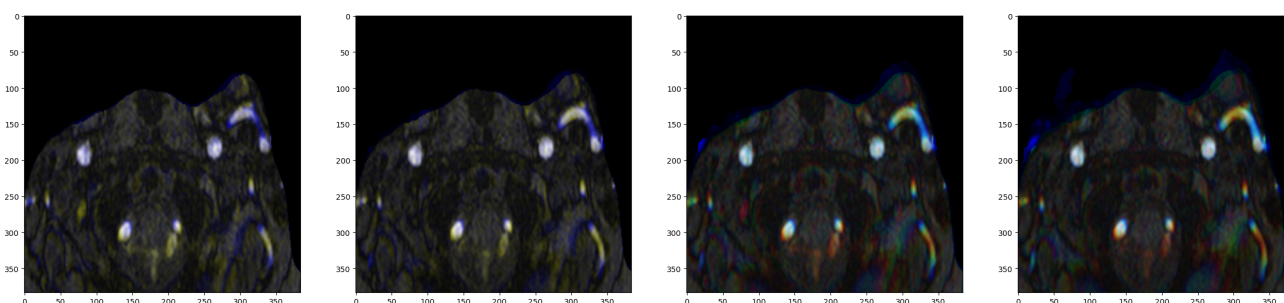
ROI Crop coordinates Model

I shifted to a ViT (dinov3) model that takes in input of equidistant 48 slices for each patient, input shape: 48x1x128x128 volume and predicts x1, x2, y1, y2 values of where the crop boundaries of segmetation mask should be at patient-level, this model was far more reliable and efficient for ROI cropping

Original:



ROI Crop:



I kept the ROI relatively large as I did not want to miss out on aneurysms, I tested with the coordinate data and doing it in this configuration let me keep 95%+ of aneurysm locations inside the cropped region

Classification Model

I took ~2.2k samples in the train_localizer csv file and all samples from negative patients which gives me a dataset of:

- 545k samples with ~2.2K positive samples and rest negative samples (1 in 250 positive),
- There are 14 labels in the dataset, which keeps it that model needs to predict 1/1700 values to be 1, rest 0s

I managed to find a combination of pipeline where model starts to learn even in this highly imbalance without weighted sampling (weighted sampling did not help).

My entire solution is this classification model and here's how I improve it's score (very simple once baseline starts learning)

My baseline hyperparameters:

- Model: CoaT-Lite-Medium
- Optimizer: AdamW | Learning-Rate: 1e-4 cosine annealing down to ~3e-5
- Augmentations: HFlip, VFlip, no TTA
- Image size: 384x384
- Global Batch-Size: 192

Approach	CV	Public/Private Leaderboard
CoaT-Lite-Medium on ROI Crops	0.805	0.78/0.78
+ Rotate (-25, 25)	0.83	0.8/0.81
+ 2.5D (-2, 0, +2)	0.86	0.86/0.83
+ Soft Pseudo-Distill on remaining 500k potentially positive samples	0.89	0.86/0.84
Ensembling with MaxViT + CoaT	0.896	0.87/0.84

On inference, the model predicts on every image for each patient, and final predictions are simply max aggregated

Thank you for reading!

Inference Code [here](#)

Training Code [here](#)

Model Weights [here](#)

Supplemental Q&A

shiba-inu · 2025-10-20T16:08:57.830000

Congrats!! Thank you very much for providing such a simple solution!

I have a question about Training.

When I was training my model, I used BCELoss and FocalLoss, but the training loss was quite unstable no matter how much I adjusted the learning rate or scheduler.

Did you experience the same issue in your training pipeline?

Or did you confirm that the loss converged smoothly?

IAmParadox · 2025-10-20T16:25:32.570000

Hey [@harshitsheoran](#) congrats on GM!, Your solution is very simple and elegant. I am trying to replicate your solution by taking some insights from your [prelims-code](#). But it doesn't seem to have the training

pipeline for cropping (maybe I just missed it). I am curious to what IoU score you got when doing such cropping using dinoV3? I am getting around ~0.65 on val

Harshit Sheoran · 2025-10-20T17:08:44.100000

Hi, Glad you liked my solution

Your score is near 0.7 which indicates that the model is not learning to predict the aneurism, it is likely that model has been picking up on other clues like weight imbalance between several modalities or other clues in the image to produce a score near 0.7, once your model really starts learning the competition target (0.78+ in my experiments), the training/validation loss/score metrics gets stable

I did face heavy unstability issues when my score was lower than 0.7

Yes, in my most recent work of scores 0.8+ the loss converges smoothly

Harshit Sheoran · 2025-10-20T17:20:05.293000

Hi [@iamparadox](#),

Congratulations to you too, Happy that you liked my solution and are excited to dive in

For cropping, I am training a ViT-Small-Plus DinoV3 pretrained model, you can use any model really..., to take input a volume (48, 1, 128, 128) and predict (x1, x2, y1, y2) for where the volume should be cropped

x1, x2, y1, y2 are calculated by taking xmin, xmax, ymin, ymax around the 3D mask volume

The metric I am using is MAE, I get an MAE of 0.027 (xy values normalized to 0 and 1)

for cropping I take `image[0.9*ymin:1.1*ymax, 0.9*xmin:1.1*xmax]`

I am sharing just about all of my code to reproduce everything:

<https://www.kaggle.com/datasets/harshitsheoran/rsna2025-training-code>

And all the raw logs, OOF outputs, and model weights:

<https://www.kaggle.com/datasets/harshitsheoran/rsna2025-raw-models/data>

shiba-inu · 2025-10-21T00:57:32.513000

Thank you for your reply! So, if the training is going well, the loss stabilizes and converges — that's reassuring to hear.

By the way, I have a question: when the loss became stable (from 0.7 to 0.78), did you do anything in particular to achieve that? For example, did you add class weights or something similar?

Harshit Sheoran · 2025-10-21T03:42:04.980000

For me, it was data cleaning, I take 2k positive samples and 500k negative samples, it was hard for the model to start learning immediately

with ROI crop I think my score went from 0.7 to 0.8 (baseline)

Tom · 2025-10-21T05:04:41.740000

Hi @harshitsheoran, congratulations on your 4th-place finish! Your approach is simple yet very effective. I'm really impressed that you managed to develop everything within just 14 days, it must have taken quite some effort to verify the effectiveness of each component.

I have two questions:

- it seems the 2.5D contributed the most to your approach. Our team didn't experiment with different sampling ranges like $(-2, 0, 2)$ or wider. How did the sampling range affect your results? Have you tried other ranges?
- Also, I'm curious, what was your motivation for choosing ROI classification instead of pursuing a more refined object detection approach? Our team observes doing localization + classification task have better result then only doing classification.

Thank you!

Harshit Sheoran · 2025-10-21T06:43:55.047000

Hi Tom,

Thank you for appreciating my work

14 days meant limited experiments on my part and I wish that I could have refined every method more much

I had tried with $[-1, 0, 1]$ earlier but at that time my sampling was positive-overweighted and it did not improve (neither did $[-2, 0, 2]$), I did not check on $[-1, 0, 1]$ when the model reached 0.8 AUC

ROI classification is simpler, yet effective, it's a tried and tested technique that I have developed many times, being short on time and thinking about complexity that an object detection approach would have both to train and infer, direct ROI coordinate prediction was just a couple hours of work from idea to finish.